

Supplement 3: Additional Figures

Two survey galleries referred to from the main paper. Each shows a family of runs at a glance rather than carrying an argument on its own: the first sweeps the rotation offsets that Part I classifies, the second shows the spacetime fields behind the diversity mechanisms of Part II. They are collected here so the main narrative is not interrupted by four full pages of plates. Figures S3.3–S3.7 are measurement figures relocated from the main paper’s ninth draft: the rotation-population curve, the two-4-cycle contrast, the diversity-control schedule, and the two damage-sweep panels whose result the main text’s placement figure carries.

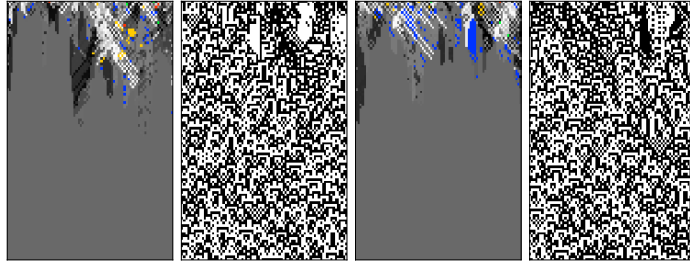
A family of operators, ranked by cycle structure (a · negative offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset -4

gcd 4 · four 2-cycles

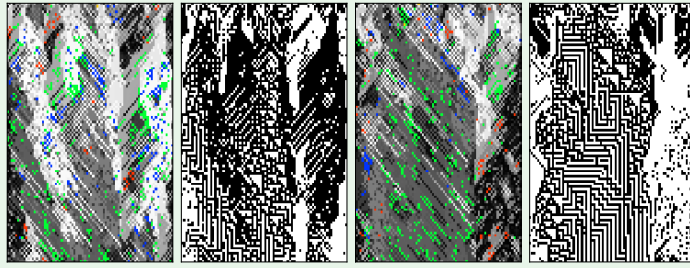
collapses ~t45



offset -3

gcd 1 · one 8-cycle

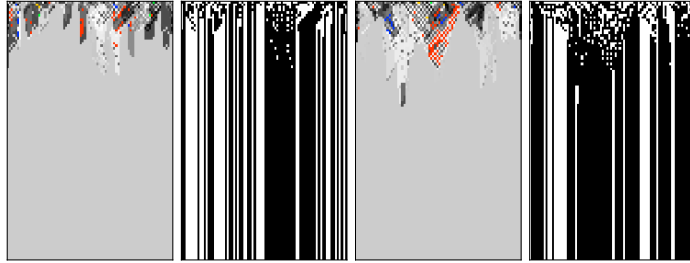
no collapse (~t100)



offset -2

gcd 2 · two 4-cycles

collapses ~t28



offset -1

gcd 1 · one 8-cycle

no collapse (~t100)

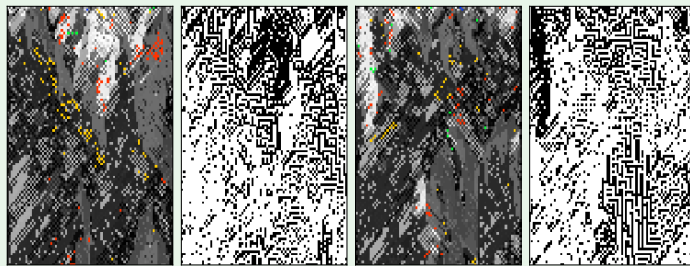


Figure S3.1

(a) Negative offsets. The rotation subfamily, ordered by offset: each row is one wrap rotation $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$ (two seeds; genotype in 256 colours, phenotype black and white, first hundred generations). Of these all-even rotations only the single 8-cycles, $\text{gcd}(\text{offset}, 8) = 1$ —offset ± 1 and ± 3 , highlighted—sustain a high-diversity phase; offset ± 2 (gcd 2) and ± 4 (gcd 4) collapse. See Worked example 4; continued in (b).

A family of operators, ranked by cycle structure (b · positive offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

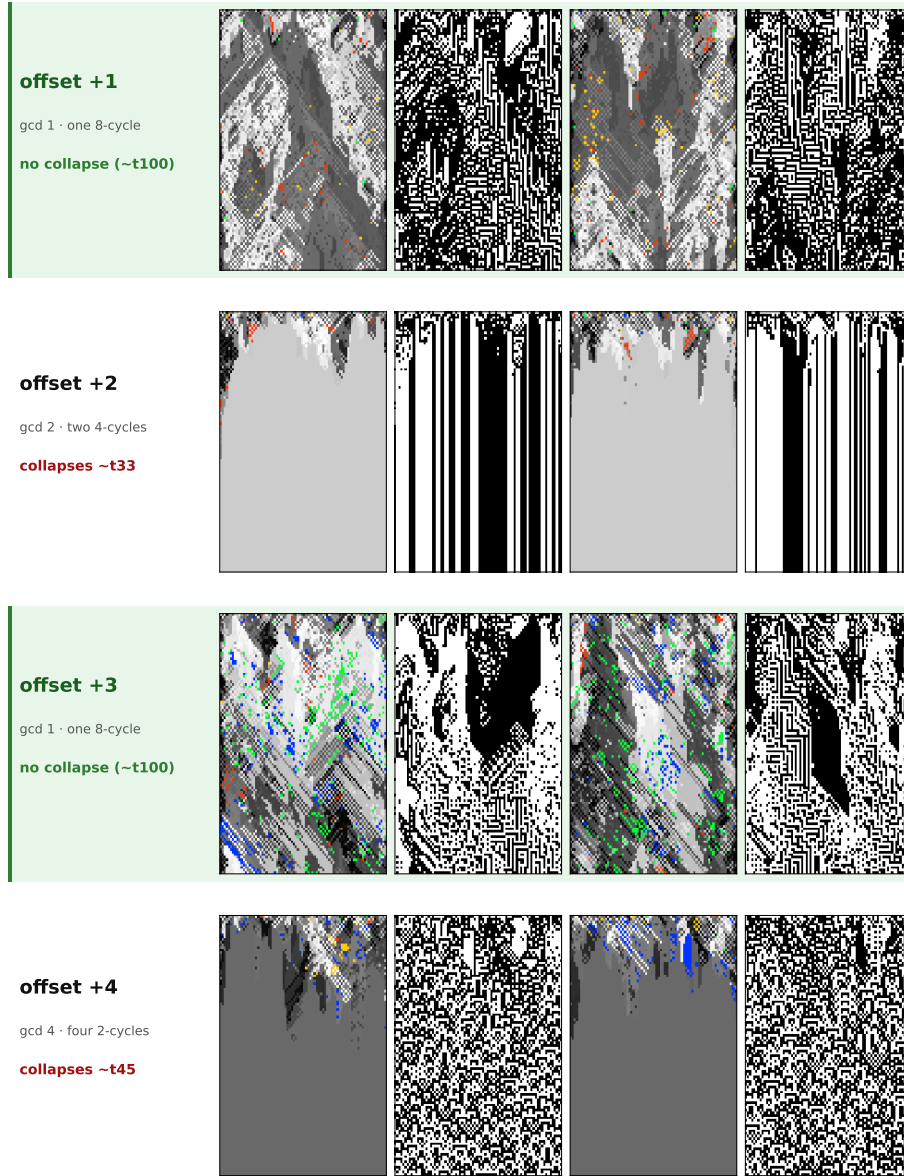


Figure S3.1

(b) Positive offsets. Continued from (a). The classification is symmetric under $d \rightarrow -d$ (verdict follows $\gcd(d, 8)$), so offset +1 sustains as offset -1 does, and offset +2 collapses as offset -2 does—inverse rotations sharing verdict but not their transients (Worked example 4). Possessing a fixed point and staying near it are different properties: the two pages visually summarise the distinction between fixed-point existence and spatial persistence.

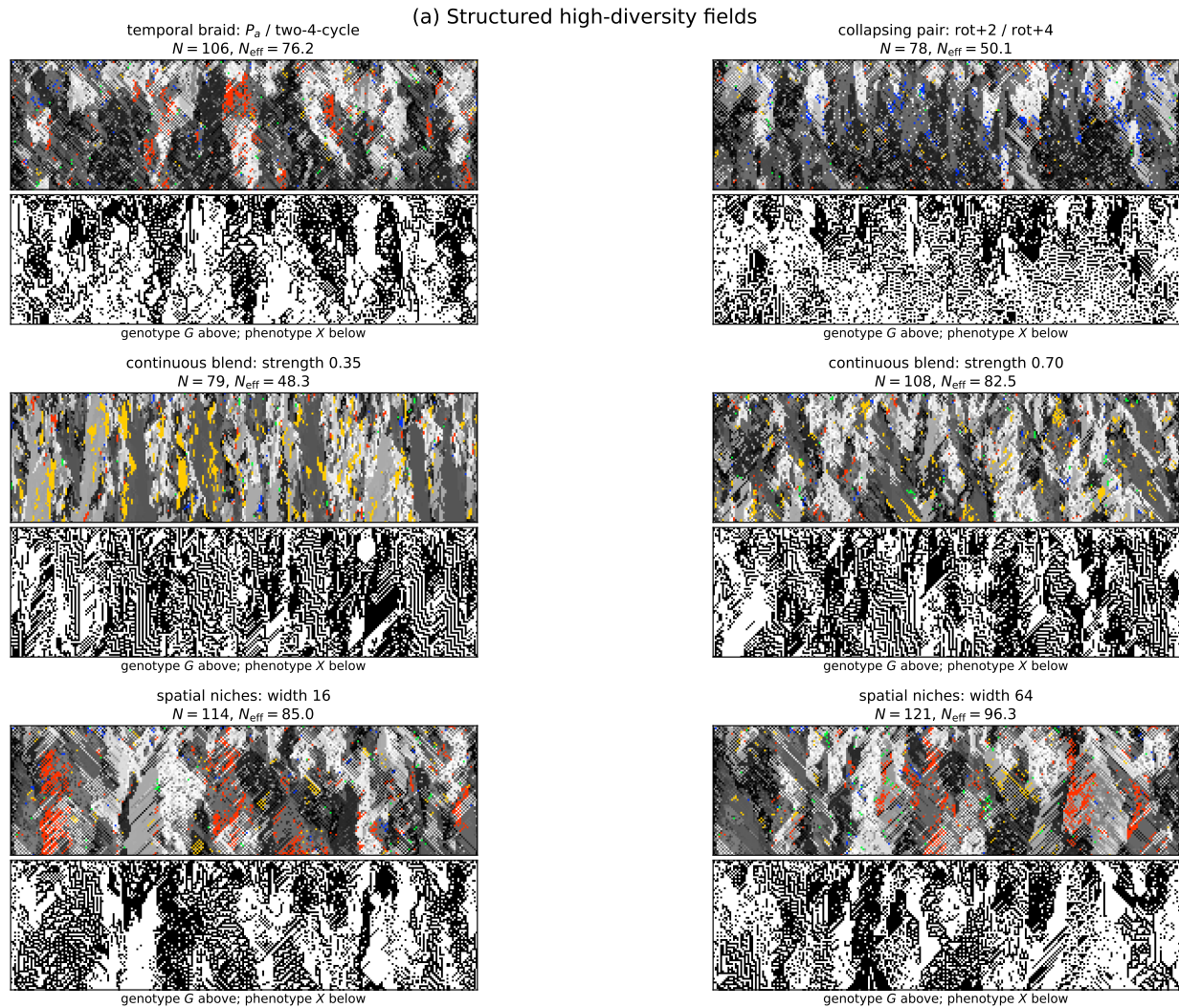


Figure S3.2

(a) Structured high-diversity fields. Six examples from four mechanism families: temporal alternation; alternation of two operators that collapse when run alone; continuous blending at two strengths; and spatial niches at two widths. Each tile stacks genotype over phenotype and reports both distinct-rule and Shannon-effective diversity at $T = 70$; full-population seed 0, $L = 256$. The fields sustain 78–121 rules and carry coherent diagonal structure at a range of scales. These examples motivated the candidate diversity coordinate, but the conservative-flow control in Figure 16 shows that diversity itself does not account for their propagation. Continued in (b).

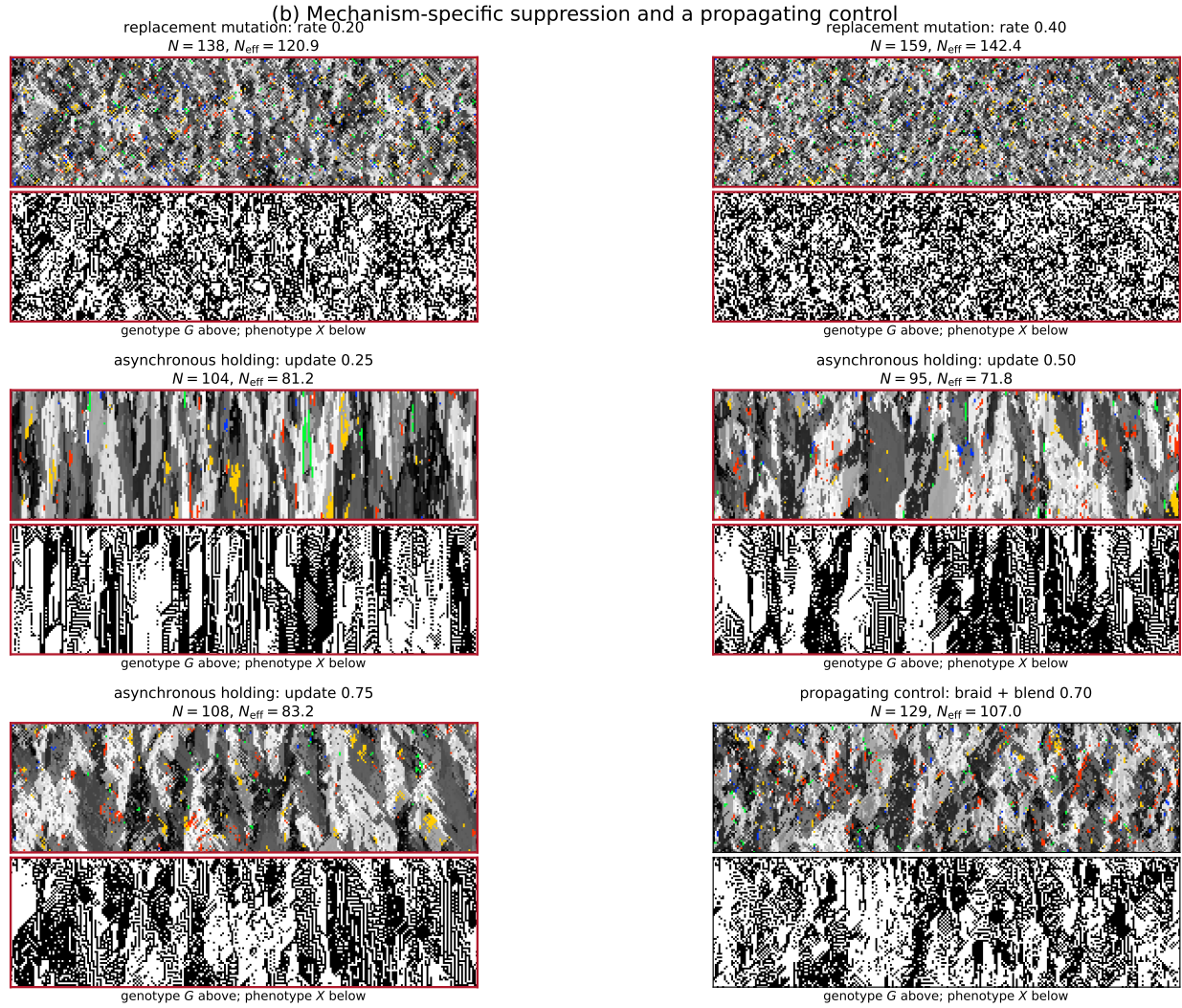
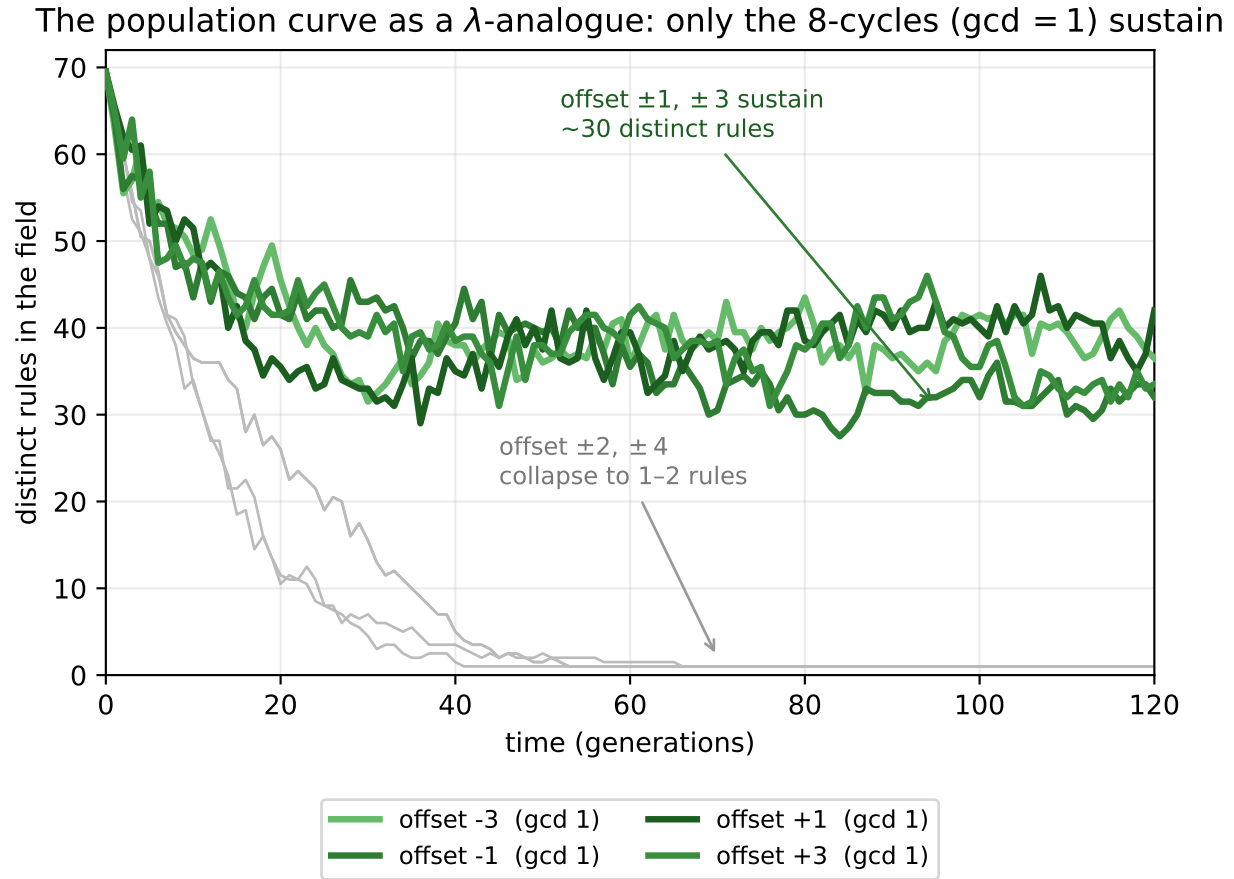


Figure S3.2

(b) *Two mechanism-specific suppressions.* Continued from (a); suppressed configurations are outlined in red. Replacement mutation at rates 0.20 and 0.40 turns the genotype into progressively finer, unstructured speckle. Asynchronous holding at update fractions 0.25, 0.50, and 0.75 instead produces vertical persistence whose strength changes continuously with the dial. The unoutlined braid-plus-blend control at bottom right propagates (3.88 cells) and recovers the diagonal texture of (a). The expanded population makes the distinction visible as a family resemblance rather than a single selected contrast: these are genuine failure modes of their respective mechanisms, not evidence for suppression by high diversity.

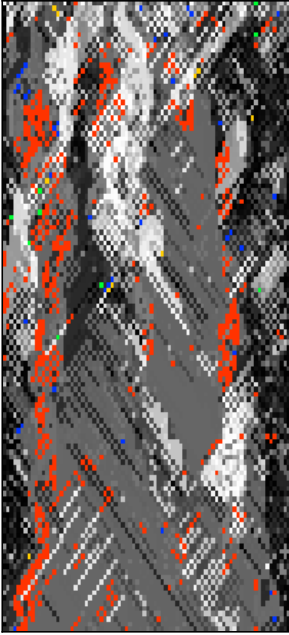
**Figure S3.3**

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70 ; the four 8-cycles, $\gcd(\text{offset}, 8) = 1$ (offset ± 1 and ± 3 , green), keep a large diverse population, while offset ± 2 and ± 4 (grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure S3.1 of this supplement. The main paper's parity-dichotomy figure gives the corresponding without-blend comparison for offsets $+1$ and $+2$: the former remains active at lower diversity over that run, while the latter settles onto its four fixed rules.

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[6 7 0 2 1 4 3 5] — **two 4-cycles**

sustains (live / live)
genotype

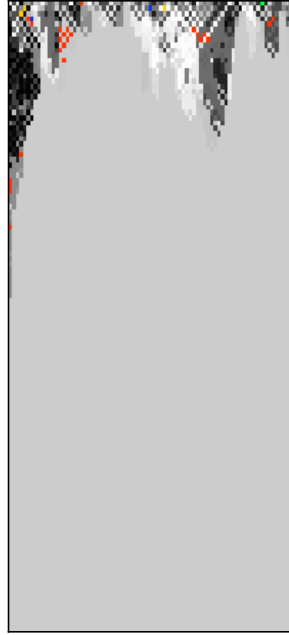


phenotype



[2 3 4 5 6 7 0 1] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



phenotype

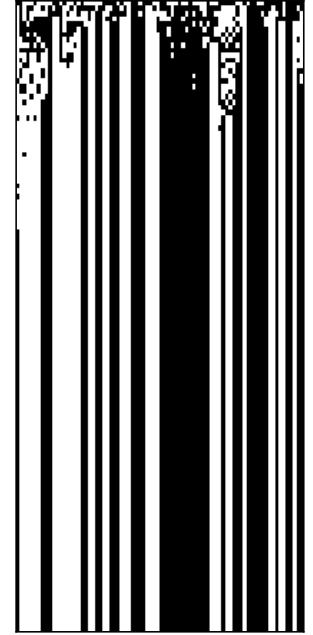


Figure S3.4

***Cycle structure does not determine aliveness.** Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left, [6 7 0 2 1 4 3 5] sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the historical offset+2 operator (its conjugate). Right, the Wolfram-order rotation offset+2 = [2 3 4 5 6 7 0 1] has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.*

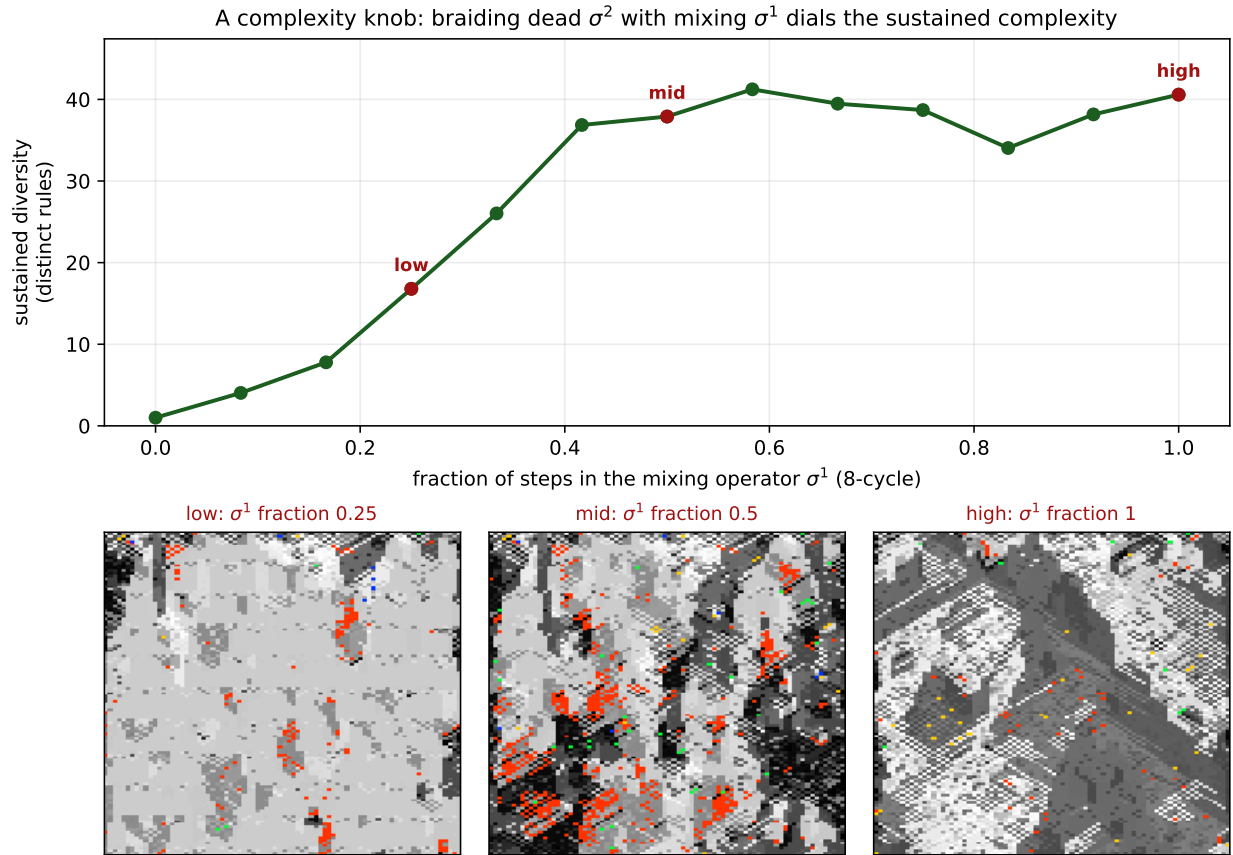


Figure S3.5

A diversity-control schedule. Periodic 12-step schedules mix the sustaining offset+1 operator σ^1 with the collapsing offset+2 operator σ^2 . Mean late-window distinct-rule diversity (five seeds) rises from 1.0 with no σ^1 steps to a high-diversity plateau once its fraction reaches roughly 0.4. The three fields show fractions 0.25, 0.50, and 1.0 at the same representative seed.

A single phenotype-bit flip spreads farther in the river than in the feedback-cut control
(same seed, same RNG tape, same construction; single-seed sweep over all 80 sites, $L=80$)

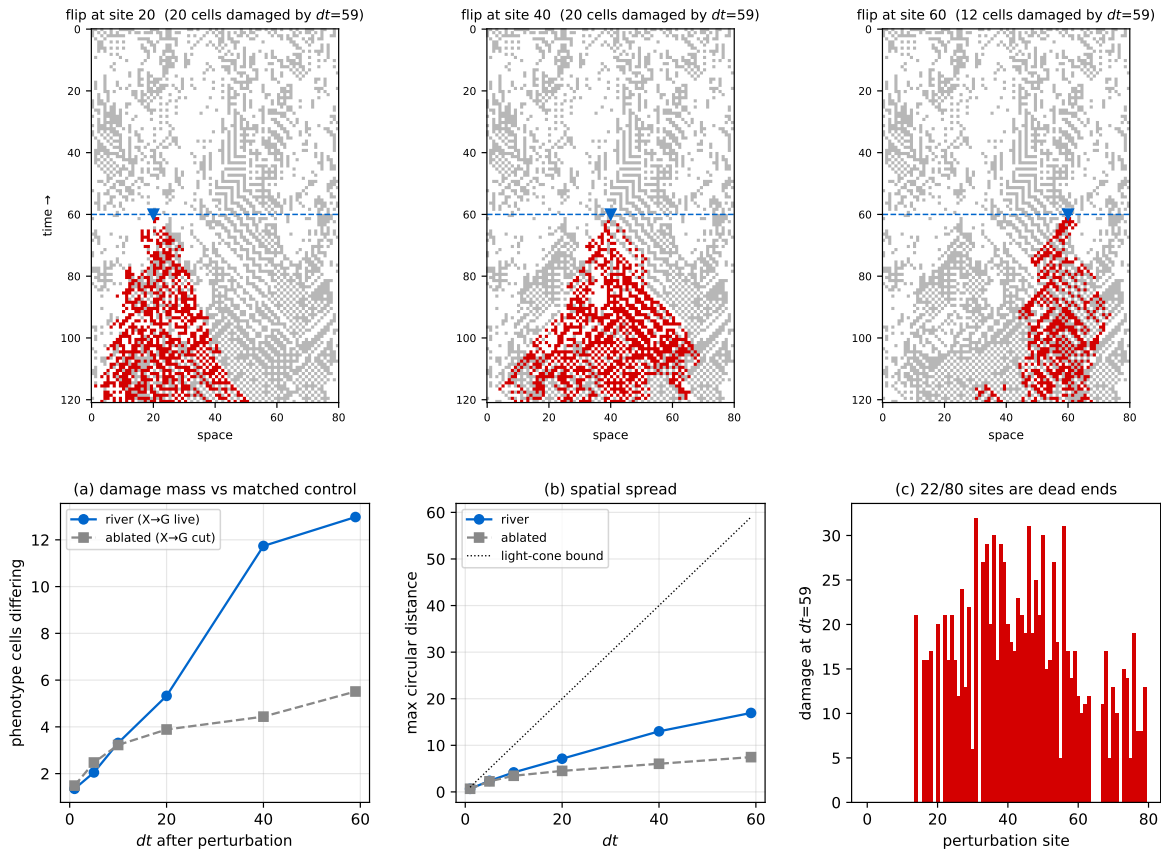


Figure S3.6

Damage from a single phenotype flip. Live construction against its matched control, $L = 80$, $t^* = 60$, swept over all 80 sites; mass and spread at a sequence of horizons dt .

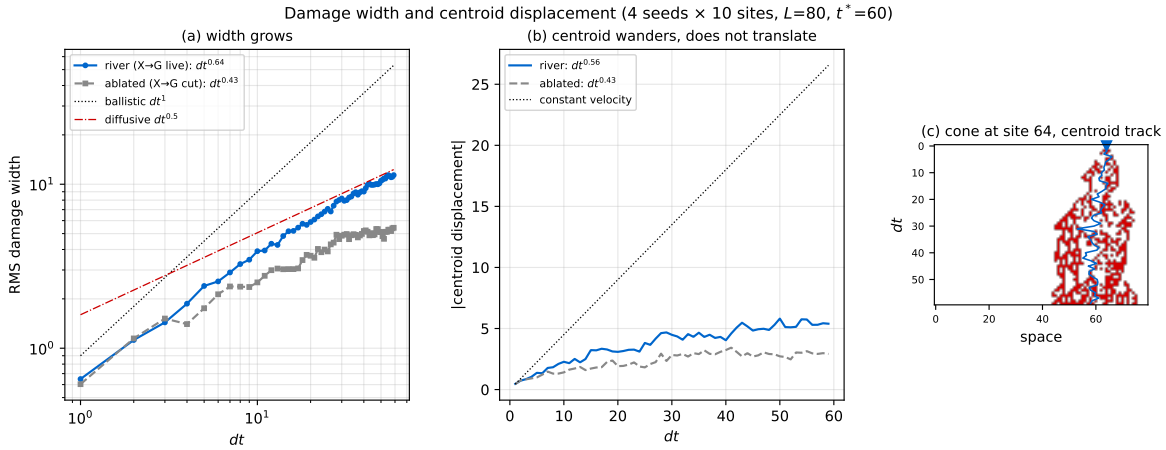


Figure S3.7

Damage width and centroid displacement. (a) damage width against dt , log-log, with ballistic and diffusive references; (b) centroid displacement; (c) a single cone with the centroid track overlaid. The packet disperses while its centroid stays near the perturbation site. We do not read this as excluding particle-like transport: under the same protocol rule 110, whose gliders support universal computation, gives $dt^{0.68}$ and $dt^{0.73}$, so a random single-site perturbation does not produce a fixed-width translating packet even where gliders certainly exist.